

Project 13 — Intelligent Sales Rep Assistant

AI-Forge 2026 Capstone Project Detailed Implementation Guide

1. Introduction

The Intelligent Sales Rep Assistant is an AI-powered Sales Technology application designed to boost the productivity of sales representatives by automating administrative tasks and providing real-time insights. The system processes and listens to sales calls, extracts commercial intelligence, automatically updates CRM records, identifies upselling opportunities, drafts follow-up emails, and provides actionable performance feedback to the sales representative.

2. Project Objectives

- Automatically transcribe live or recorded sales conversations in real-time.
- Extract key sales metadata and update CRM fields without manual data entry.
- Analyze conversation dynamics to spot buying signals, cross-sell, and upsell opportunities.
- Generate hyper-personalized, contextual follow-up emails based on call outcomes.
- Provide objective coaching feedback to sales professionals to improve performance.
- Build a production-grade AI workflow following AI-Forge guidelines.

3. Core Features

- **Call Transcription Feature:** Transcribes sales calls accurately in real-time or from audio recordings, capturing speaker sentiments and timestamps.
- **CRM Automation Feature:** Extracts structured key information from the call transcription and programmatically updates relevant pipelines and fields in the CRM.
- **Opportunity Spotting Feature:** Analyzes customer conversations for strong buying signals, explicitly flagging potential upsell or cross-sell options.
- **Email Generation Feature:** Drafts personalized, high-conversion follow-up emails tailored specifically to the call's exact content and finalized outcome.
- **Sales Coach Feature:** Provides analytical, constructive feedback to the sales representative regarding their communication performance and closing techniques based on call metrics.

4. AI Concepts Used

- **Natural Language Processing (NLP) & Speech-to-Text:** Used to process unstructured audio streams into text, understand conversational context, and perform entity

classification.

- **Sales Intelligence AI:** Specialized AI modeling focused on commercial terminology, revenue operations, objection handling, and buying signal identification.
- **Sequential Pipeline Processing:** A deterministic AI pipeline where sequential stages depend directly on the high-quality outputs generated by previous modules.
- **Conversation Understanding:** The system deeply interprets spoken structures, customer intent, implicit obligations, and temporal action items.

5. Recommended System Architecture

Frontend (React) → Backend API (Python) → Audio/Stream Upload Service → Real-Time Speech-to-Text Pipeline → Transcript Chunking → Sequential AI Pipelines → CRM Automation → Opportunity Spotting → Email Generation → Sales Coach Module → Aggregation & Synchronization Layer → Supabase Database → Sales Insights & Report Dashboard

6. Recommended Technology Stack

Layer	Technology
Frontend	React
Backend	Python
AI Framework	LangChain
Pipeline Orchestration	LangGraph
Workflow Automation	n8n
Database & Auth	Supabase
Speech-to-Text Engine	OpenAI Whisper / AssemblyAI
LLM API	GPT-4 / Claude
Vector Database	Pinecone / ChromaDB

7. Step-by-Step Development Plan

1. **Phase 1 — Requirement Analysis:** Understand conversational intelligence requirements, CRM schema formats, and expected final sales metrics.
2. **Phase 2 — Build MVP:** Implement basic audio processing, speech-to-text integration, and fundamental summary workflow.
3. **Phase 3 — Transcription Feature:** Develop optimized speech pipelines with speaker diarization and time-stamping.
4. **Phase 4 — CRM Automation:** Build structured data extraction logic and prompt templates to map details into mock CRM schemas.
5. **Phase 5 — Opportunity Spotting:** Implement intent classification models and buying signal scoring algorithms.

6. **Phase 6 — Email & Coaching Modules:** Author generative prompts for email personalization and establish standard rubric scoring for sales coaching.
7. **Phase 7 — Database Integration:** Store transcriptions, structured metrics, and performance analytics securely in Supabase.
8. **Phase 8 — UI Dashboard:** Develop a frontend representative dashboard displaying active pipelines, audio uploads, and insights visualization.
9. **Phase 9 — Testing & Validation:** Perform end-to-end user acceptance testing using diverse mock sales calls and optimize LLM prompts.
10. **Phase 10 — Final Demonstration:** Prepare sequential pipeline architecture explanation, workflow demonstration, and an engineering code walkthrough.

8. Workflow Process

1. The sales representative conducts or uploads a recorded sales call.
2. The Speech-to-Text engine processes the audio file and establishes real-time transcription.
3. The transcription text is parsed and divided logically into readable conversational chunks.
4. The sequential pipeline initiates processing on the curated transcript data.
5. The CRM automation component extracts structured information and fills necessary fields.
6. The opportunity spotting agent scans conversation logs to detect upsell or cross-sell opportunities.
7. The email generation feature builds a tailored context-aware draft based on the call's outcome.
8. The sales coach feature analyzes the rep's overall performance and compiles actionable tips.
9. The aggregation layer combines all extracted sales metrics and performance insights.
10. The finalized insights, email draft, and updates are saved in Supabase and pushed live to the dashboard.

9. Expected Final Deliverables

- Working AI-powered Intelligent Sales Rep Assistant application.
- Frontend sales dashboard showcasing pipeline metrics, transcripts, and analytics visualization.
- Sequential AI pipeline workflow implementation.
- Database integration with Supabase for data resilience and logging.
- System and pipeline architecture diagrams.
- Comprehensive code walkthrough and developer documentation.
- End-to-end demonstration featuring an analysis of a simulated sales interaction.

10. Best Practices & Recommendations

- Use structured JSON schemas from LLM outputs to guarantee programmatic CRM syncs.
- Implement rolling semantic text window chunking to maintain continuous conversational context.
- Maintain comprehensive call transcript and generated insight logs for strict quality

- auditability.
- Incorporate accuracy and confidence scores for highlighted upsell or cross-sell recommendations.
- Keep the codebase completely modular to scale for future CRM tool integrations (e.g., Salesforce, HubSpot).
- Focus heavily on production-quality prompt engineering and robust runtime execution handling.

11. Conclusion

The Intelligent Sales Rep Assistant represents a production-grade Sales Tech AI system that masterfully combines automated speech translation, contextual intent classification, and structured pipeline automation. This project aligns closely with contemporary enterprise B2B sales automation use cases and showcases advanced AI architecture engineering expected in the AI-Forge 2026 Capstone Programme.